

Heterogeneity-Driven Expansion and Synchronization Collapse in Coupled Ornstein–Uhlenbeck Systems

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Abstract

We study two diffusively coupled Ornstein–Uhlenbeck processes with unequal noise amplitudes and analyze the stationary covariance matrix $\Sigma(\kappa)$ exactly, through the continuous Lyapunov equation, as a function of the coupling strength κ . Three scalar functionals of $\Sigma(\kappa)$ are considered: the residual variance $D(\kappa) = \text{Var}(X | Y)$, the Gaussian mutual information $C(\kappa) = I(X; Y)$, and the stationary covariance-volume proxy $A(\kappa) = \det \Sigma(\kappa)$. We prove that for arbitrary positive relaxation rates and noise amplitudes, $A'(0) < 0$, so infinitesimal diffusive coupling always initially contracts the stationary covariance volume. In the equal-relaxation case, we then prove the exact threshold

$$\sup_{\kappa > 0} A(\kappa) > A(0) \iff \max\left(\frac{\sigma_1}{\sigma_2}, \frac{\sigma_2}{\sigma_1}\right) > 3 + 2\sqrt{2}.$$

When this threshold is exceeded, the enhancement occurs on a bounded interval whose two endpoints satisfy the exact reciprocal identity $\kappa_1 \kappa_2 = 1$. A second exact threshold, $2 + \sqrt{3}$, governs whether the conditional residual variance $D(\kappa)$ can exceed its uncoupled value. Thus distinct legitimate measures of relational structure acquire enhancement at different heterogeneity scales. We also prove that Gaussian coherence $C(\kappa)$ increases strictly with coupling, while in the strong-coupling limit $\rho \rightarrow 1$, $D(\kappa) \rightarrow 0$, $A(\kappa) \rightarrow 0$, with explicit asymptotic rates.

This version adds three things to the August 2026 version. First, the two thresholds are shown to share a single reciprocal structure: both are roots of $x^2 - cx + 1 = 0$, with $c = 6$ for the volume threshold and $c = 4$ for the residual threshold, and at the coupling where the volume enhancement region closes the eigenvalue ratio of Σ equals $(2 + \sqrt{3})^2$ exactly. Second, the central algebraic statements have been machine-checked in Lean 4 with mathlib. Third, several statements of the first version have been corrected or sharpened, in particular the description of the normal-mode reduction, in which heterogeneous noise makes the two normal modes *correlated* rather than independent. The results are specific to this linear-Gaussian model and are not claimed to establish a general principle for nonlinear, biological, cognitive, or social systems.

0. Status of this version

Reconstruction notice. The LaTeX source of version 1 was not preserved. This document was reconstructed from the deposited PDF of version 1 and then revised. Every displayed equation of version 1 was transcribed and re-verified symbolically before being carried over; the surrounding prose is a reconstruction and differs from version 1 in wording and in equation and section numbering. Readers wanting the exact text of version 1 should consult that version of the record, which remains available and is not superseded as a record of what was written in August 2026.

What changed. A full list is in `CHANGELOG.md` accompanying this version. The substantive changes are:

1. A new Section 11 on the reciprocal structure shared by the two thresholds and on a cross-threshold identity relating them at the merger coupling.
2. A new Section 15 reporting a machine-checked formalization of the central algebraic statements in Lean 4 with mathlib, together with an explicit statement of which results of this paper are *not* formalized.
3. A correction, in Section 7, to the treatment of the normal-mode decomposition: the symmetric and antisymmetric modes have uncorrelated forcing only in the homogeneous case $\sigma_1 = \sigma_2$. For $\sigma_1 \neq \sigma_2$ they are driven by correlated noise, and that correlation is exactly what produces the enhancement regime.
4. A sharpening of terminology around the word *synchronization* (Section 14, Remark 5).
5. An explicit distinction between the amplitude ratio r and the variance ratio $t = r^2$, and between their respective reciprocal invariants (Section 5.3).
6. Strengthened limitations (Section 18).
7. A one-paragraph pointer to a companion formalization of a general two-mode reduction that contains the present volume threshold as a special case (Section 20).
8. A fuller statement of AI assistance (Acknowledgments).

No result of version 1 is retracted. The thresholds $3 + 2\sqrt{2}$ and $2 + \sqrt{3}$, the identity $\kappa_1\kappa_2 = 1$, the sign of $A'(0)$, the monotonicity of C , and the strong-coupling asymptotics all stand as stated, and several of them now stand as machine-checked statements of exact real algebra.

1 Introduction

Coupling two dynamical systems can either enlarge or reduce the range of configurations occupied by the joint system. At one limit, weakly coupled systems remain nearly independent. At the opposite limit, sufficiently strong diffusive coupling can drive the components toward a common synchronization manifold, reducing their effectively independent degrees of freedom.

A natural intermediate question is therefore whether coupling can increase the joint configuration volume before synchronization eventually reduces it.

The difficulty is that terms such as “coherence,” “distinction,” “richness,” and “accessibility” can refer to mathematically different objects. In the present paper we avoid treating them as interchangeable. Instead, we analyze a minimal model in which several such quantities can be calculated exactly: two diffusively coupled Ornstein–Uhlenbeck (OU) processes with independent additive Gaussian noise [21, 10].

Because the system is linear and Gaussian, its stationary dynamics are fully characterized by a 2×2 covariance matrix $\Sigma(\kappa)$. This permits three related but distinct quantities to be studied explicitly:

$$D(\kappa) = \text{Var}(X \mid Y), \tag{1}$$

$$C(\kappa) = I(X; Y), \tag{2}$$

$$A(\kappa) = \det \Sigma(\kappa). \tag{3}$$

The principal result is an exact threshold for the last quantity. For equal intrinsic relaxation rates, the stationary covariance volume can exceed its uncoupled value at some finite coupling strength if and only if the noise-amplitude ratio exceeds $3 + 2\sqrt{2}$.

A second exact threshold, $2 + \sqrt{3}$, controls whether the residual variance $D(\kappa)$ can exceed its uncoupled value. Thus even within this simplest solvable setting, different legitimate measures of relational structure do not acquire enhancement at the same parameter value.

The results should be read narrowly. We establish an exact property of a two-dimensional linear stochastic model. We do not infer from it a general law of heterogeneous interaction.

2 Relation to Existing Work

The present model lies at the intersection of several established literatures.

Synchronization. The reduction of independent degrees of freedom under strong coupling is part of the broad theory of synchronization of coupled dynamical systems [19]. For networks of identical oscillators, the master-stability framework of Pecora and Carroll separates longitudinal and transverse perturbation modes relative to a synchronization manifold [18]. Generalized synchronization extends this idea to functional relations between nonidentical systems [20, 14, 1]. As emphasized in Remark 5 below, the present paper borrows the *geometry* of that decomposition — a longitudinal and a transverse mode relative to $x = y$ — but not its stability-theoretic content: our system is linear and always has a unique stationary distribution, so no master stability function is involved and nothing here is synchronization in the sense of those works.

Amplitude and oscillation death. Strong coupling can also suppress oscillatory dynamics entirely in nonlinear oscillator systems [2]. That mechanism is distinct from the present one: our OU system possesses no limit cycle, and the strong-coupling collapse concerns the stationary covariance ellipse approaching a lower-dimensional line rather than the disappearance of an oscillation.

Heterogeneous OU processes. Stationary covariance matrices of coupled heterogeneous OU processes have been studied directly through Sylvester–Lyapunov equations. Barucca investigated localization phenomena in covariance matrices of heterogeneous OU processes [4], and Ferreira, Metz, and Barucca subsequently developed a random-matrix ensemble for covariance matrices of OU systems with heterogeneous temperatures [8]. Godrèche and Luck give exact stationary results for coupled linear Langevin systems driven at two temperatures [11], and the Brownian gyrator literature [9, 7] studies the same two-temperature structure in a linear planar model. These works provide the closest mathematical setting to the present calculation, and the modal solution used in Section 7 is standard in them.

We did not identify in the literature examined the specific two-node enhancement thresholds $3 + 2\sqrt{2}$ and $2 + \sqrt{3}$, nor the cross-threshold identity of Section 11. No claim of priority is made beyond the scope of that search; a fuller account of what was searched, and of which ingredients here are standard, is kept with the formalization repository.

Information theory. For jointly Gaussian variables, mutual information, conditional variance, covariance determinants, and differential entropy have standard closed forms [5, 15]. We use these exact identities directly. The determinant $\det \Sigma$ as a scalar summary of a covariance matrix is Wilks’s generalized variance [23].

Viability language. Constraint-based formulations of persistence are central to viability theory [3]. A set of coupling strengths satisfying simultaneous conditions on D , C , and A can be viewed informally as a feasible parameter region. We do not, however, claim that the parameter-space construction used later is itself a viability kernel in the strict state-space sense.

Synergetics. The elimination of fast transverse degrees of freedom under strong coupling has a conceptual relation to slaving and dimension reduction in synergetics [12]. The analogy is structural rather than an identification of mechanisms.

3 Coupled Ornstein–Uhlenbeck Model

Consider

$$dx = -(a_1x + \kappa(y - x)) dt + \sigma_1 dW_1, \quad (4)$$

$$dy = -(a_2y + \kappa(x - y)) dt + \sigma_2 dW_2, \quad (5)$$

with $a_1, a_2 > 0$, $\sigma_1, \sigma_2 > 0$, $\kappa \geq 0$, and independent standard Brownian motions W_1, W_2 .

Writing $Z = (x, y)^\top$, the dynamics become

$$dZ = MZ dt + B dW, \quad (6)$$

where

$$M = \begin{pmatrix} -(a_1 + \kappa) & \kappa \\ \kappa & -(a_2 + \kappa) \end{pmatrix}, \quad B = \begin{pmatrix} \sigma_1 & 0 \\ 0 & \sigma_2 \end{pmatrix}. \quad (7)$$

The noise covariance is $Q = BB^\top = \text{diag}(\sigma_1^2, \sigma_2^2)$. The trace and determinant of the drift matrix are

$$\text{tr } M = -(a_1 + a_2 + 2\kappa) < 0, \quad \det M = a_1a_2 + \kappa(a_1 + a_2) > 0,$$

so M is Hurwitz for all allowed parameter values.

Proposition 1 (Stationary Gaussian state). *For every $a_1, a_2, \sigma_1, \sigma_2 > 0$ and $\kappa \geq 0$, the system possesses a unique zero-mean stationary Gaussian distribution whose covariance matrix Σ solves*

$$M\Sigma + \Sigma M^\top = -Q. \quad (8)$$

This is the standard stationary covariance relation for a linear stochastic system with Hurwitz drift matrix [10, 22].

4 Exact Stationary Covariance

Write

$$\Sigma = \begin{pmatrix} \Sigma_{xx} & \Sigma_{xy} \\ \Sigma_{xy} & \Sigma_{yy} \end{pmatrix}.$$

Solving Eq. (8) gives the following exact expressions.

Proposition 2 (Exact stationary covariance). *Define*

$$\Delta = a_1^2a_2 + a_1^2\kappa + a_1a_2^2 + 4a_1a_2\kappa + 2a_1\kappa^2 + a_2^2\kappa + 2a_2\kappa^2. \quad (9)$$

Then

$$\Sigma_{xx} = \frac{a_1 a_2 \sigma_1^2 + a_1 \kappa \sigma_1^2 + a_2^2 \sigma_1^2 + 3a_2 \kappa \sigma_1^2 + \kappa^2 \sigma_1^2 + \kappa^2 \sigma_2^2}{2\Delta}, \quad (10)$$

$$\Sigma_{yy} = \frac{a_1^2 \sigma_2^2 + a_1 a_2 \sigma_2^2 + 3a_1 \kappa \sigma_2^2 + a_2 \kappa \sigma_2^2 + \kappa^2 \sigma_1^2 + \kappa^2 \sigma_2^2}{2\Delta}, \quad (11)$$

$$\Sigma_{xy} = \frac{\kappa(a_1 \sigma_2^2 + a_2 \sigma_1^2 + \kappa \sigma_1^2 + \kappa \sigma_2^2)}{2\Delta}. \quad (12)$$

Consequently,

$$\begin{aligned} A(\kappa) := \det \Sigma(\kappa) &= \frac{a_1^2 \sigma_1^2 \sigma_2^2 + 2a_1 a_2 \sigma_1^2 \sigma_2^2 + 4a_1 \kappa \sigma_1^2 \sigma_2^2}{4(a_1 + a_2 + 2\kappa)^2(a_1 a_2 + a_1 \kappa + a_2 \kappa)} \\ &\quad + \frac{a_2^2 \sigma_1^2 \sigma_2^2 + 4a_2 \kappa \sigma_1^2 \sigma_2^2 + \kappa^2 \sigma_1^4 + 2\kappa^2 \sigma_1^2 \sigma_2^2 + \kappa^2 \sigma_2^4}{4(a_1 + a_2 + 2\kappa)^2(a_1 a_2 + a_1 \kappa + a_2 \kappa)}. \end{aligned} \quad (13)$$

The analytic solution was cross-checked numerically against an independent continuous-Lyapunov solver at multiple parameter points. The maximum observed discrepancy was 2.8×10^{-14} . The equal-relaxation specialization of Eqs. (10)–(12) has, in addition, been machine-checked against Eq. (8) (Section 15).

5 Covariance Geometry

We define

$$D(\kappa) = \text{Var}(X | Y) = \Sigma_{xx} - \frac{\Sigma_{xy}^2}{\Sigma_{yy}} = \frac{\det \Sigma}{\Sigma_{yy}}, \quad (14)$$

and

$$C(\kappa) = I(X; Y) = -\frac{1}{2} \log(1 - \rho^2), \quad \rho = \frac{\Sigma_{xy}}{\sqrt{\Sigma_{xx} \Sigma_{yy}}}, \quad (15)$$

and finally $A(\kappa) = \det \Sigma(\kappa)$.

For a bivariate Gaussian,

$$h(X, Y) = \log(2\pi e) + \frac{1}{2} \log \det \Sigma.$$

Thus $\det \Sigma$ is monotone in the differential entropy of the stationary distribution and sets the squared area scale of its covariance ellipse. We therefore refer to A as the stationary covariance volume or stationary configuration-volume proxy; it is Wilks's generalized variance [23].

It should not be confused with: (i) state-space dimension; (ii) support of the stationary measure; (iii) control-theoretic reachability; (iv) basin-of-attraction volume; (v) number of distinct attractors.

In particular, because $B = \text{diag}(\sigma_1, \sigma_2)$ has full rank whenever $\sigma_1, \sigma_2 > 0$, the associated deterministic control system $\dot{Z} = MZ + Bu$ has direct access to both state coordinates and is controllable for every $\kappa \geq 0$. Control-theoretic reachable-set dimension therefore cannot distinguish the coupling regimes studied here.

5.1 Residual variance as orthogonal structure

For jointly Gaussian variables, $D(\kappa) = \text{Var}(X | Y)$ is the Schur complement of Σ_{yy} [15, 13]. Equivalently, let $X_{\parallel} = E[X | Y]$ be the L^2 projection of X onto the linear span generated by Y , and write $X = X_{\parallel} + X_{\perp}$. Then $E[X_{\perp} Y] = 0$ and $D(\kappa) = \text{Var}(X_{\perp})$.

Table 1: The two thresholds in amplitude and variance variables.

	r	$s(r) = r + 1/r$	$t = r^2$	$w(t) = t + 1/t$	numerical r
volume threshold r_A^*	$3 + 2\sqrt{2}$	6	$17 + 12\sqrt{2}$	34	5.82843
residual threshold r_D^*	$2 + \sqrt{3}$	4	$7 + 4\sqrt{3}$	14	3.73205

Thus the language of “orthogonal structure” has a precise interpretation here: it means residual nonredundant variance, not a literal geometric assertion that two physical systems should meet at an angle of 90° .

5.2 One covariance object, several scalar views

The quantities D , C , and A are not independent primitives:

$$\Sigma(\kappa) \longmapsto \{D(\kappa), C(\kappa), A(\kappa)\}.$$

Their different behavior therefore represents different scalar projections of a single relational covariance geometry.

5.3 Amplitude ratio versus variance ratio

Two distinct ratios appear in this paper and are easy to conflate. We fix the convention here and use it throughout.

- $r = \sigma_2/\sigma_1$ is the *noise-amplitude* ratio. Both thresholds, $3 + 2\sqrt{2}$ and $2 + \sqrt{3}$, are values of r .
- $t = r^2 = \sigma_2^2/\sigma_1^2$ is the *noise-variance* (or temperature) ratio. It is t , not r , that enters the Lyapunov equation, since $Q = \text{diag}(\sigma_1^2, \sigma_2^2)$.

Accordingly there are two reciprocal invariants, related by a Chebyshev-type identity:

$$s(r) = r + \frac{1}{r}, \quad w(t) = t + \frac{1}{t}, \quad w(r^2) = s(r)^2 - 2. \quad (16)$$

Both are invariant under the relabelling $X \leftrightarrow Y$, which sends $r \mapsto 1/r$. The numerical values attached to the two thresholds are collected in Table 1 and used in Section 11. Statements in this paper that quote a threshold as a number always quote it as a value of r ; wherever the variance ratio is meant, t or w is written explicitly.

6 Universal Weak-Coupling Contraction

Proposition 3 (Universal negative initial response). *For every $a_1, a_2, \sigma_1, \sigma_2 > 0$,*

$$A(0) = \frac{\sigma_1^2 \sigma_2^2}{4a_1 a_2}, \quad \left. \frac{dA}{d\kappa} \right|_{\kappa=0} = -\frac{\sigma_1^2 \sigma_2^2 (a_1 + a_2)}{4a_1^2 a_2^2} < 0. \quad (17)$$

Proof. Both expressions follow by evaluation and differentiation of Eq. (13). Positivity of $a_1, a_2, \sigma_1, \sigma_2$ makes the sign in Eq. (17) strictly negative. $\square$

Thus infinitesimal diffusive coupling always initially decreases the stationary covariance volume. Any later enhancement must therefore be a finite-coupling phenomenon. Note that Proposition 3 holds for *arbitrary* positive rates and amplitudes, with no equal-relaxation assumption; it is the only result in the paper of that generality.

7 Equal-Relaxation Reduction

Set $a_1 = a_2 = 1$, $\sigma_1 = 1$, $\sigma_2 = r$, $r \geq 1$. Then

$$A(\kappa) = \frac{\kappa^2 r^4 + 2\kappa^2 r^2 + \kappa^2 + 8\kappa r^2 + 4r^2}{16(\kappa + 1)^2(2\kappa + 1)}, \quad A(0) = \frac{r^2}{4}. \quad (18)$$

Proposition 4 (Exact difference formula).

$$A(\kappa) - A(0) = \frac{\kappa[\kappa(r^4 - 18r^2 + 1) - 8r^2(\kappa^2 + 1)]}{16(\kappa + 1)^2(2\kappa + 1)} = \frac{\kappa B_A(\kappa; r)}{16(\kappa + 1)^2(2\kappa + 1)}, \quad (19)$$

where

$$B_A(\kappa; r) = -8r^2\kappa^2 + (r^4 - 18r^2 + 1)\kappa - 8r^2. \quad (20)$$

For $\kappa > 0$ the denominator and the prefactor κ are positive, so the sign of $A(\kappa) - A(0)$ is exactly the sign of $B_A(\kappa; r)$. Its discriminant is

$$\Delta_A = (r^2 - 1)^2(r^4 - 34r^2 + 1). \quad (21)$$

7.1 Normal modes and correlated forcing

It is worth recording why the enhancement regime exists at all, since the mechanism is easy to state in the drift eigenbasis and was described incorrectly in version 1.

For $a_1 = a_2 = 1$ the drift matrix of Eq. (7) is $M = -(I + \kappa L)$ with $L = \begin{pmatrix} 1 & -1 \\ -1 & 1 \end{pmatrix}$ the graph Laplacian of the single edge. Its eigenvectors are the symmetric and antisymmetric modes

$$u = \frac{x + y}{\sqrt{2}}, \quad v = \frac{x - y}{\sqrt{2}},$$

with decay rates

$$D_1 = 1 \quad (\text{longitudinal, along } x = y), \quad D_2 = 1 + 2\kappa \quad (\text{transverse}).$$

The drift is therefore diagonal in (u, v) for every κ . *The forcing is not.* Rotating $Q = \text{diag}(1, r^2)$ into the same basis gives

$$\tilde{Q} = \frac{1}{2} \begin{pmatrix} 1 + r^2 & 1 - r^2 \\ 1 - r^2 & 1 + r^2 \end{pmatrix}, \quad (22)$$

whose off-diagonal entry vanishes if and only if $r = 1$. Heterogeneous noise thus drives the two normal modes with *correlated* noise; the modes are uncoupled in the drift but not in the forcing, and the stationary covariance does not diagonalize in this basis. Solving the Lyapunov equation mode by mode, $\tilde{\Sigma}_{ij} = \tilde{Q}_{ij}/(D_i + D_j)$, gives

$$\tilde{\Sigma}_{11} = \frac{1 + r^2}{4}, \quad \tilde{\Sigma}_{22} = \frac{1 + r^2}{4(1 + 2\kappa)}, \quad \tilde{\Sigma}_{12} = \frac{1 - r^2}{4(1 + \kappa)},$$

and hence the compact form

$$A(\kappa) = \det \tilde{\Sigma} = \frac{(1 + r^2)^2}{16(1 + 2\kappa)} - \frac{(1 - r^2)^2}{16(1 + \kappa)^2}, \quad (23)$$

which is algebraically identical to Eq. (18).

Equation (23) makes the competition explicit. The first term is the product of the two modal variances and decays like $1/(2\kappa)$: coupling always shrinks the transverse mode. The second term is the squared modal correlation, present only when $r \neq 1$, and it decays faster, like $1/\kappa^2$. At small κ the first term dominates and A falls, which is Proposition 3. If the correlation term is large enough at $\kappa = 0$ — that is, if r is large enough — its faster decay can outrun the shrinking of the first term over a bounded window, and A rises above $A(0)$ there. The threshold $3 + 2\sqrt{2}$ is the exact point at which that window opens.

Remark 1. Version 1 did not contain Eq. (23) and, in describing the strong-coupling limit, spoke of the normal modes as though they were independently driven. That is correct only for $\sigma_1 = \sigma_2$, in which case there is no enhancement regime at all. The correction does not change any stated result: Eq. (23) and Eq. (18) are the same function.

8 Exact Covariance-Volume Threshold

Theorem 1 (Exact heterogeneity threshold). *For equal relaxation rates,*

$$\sup_{\kappa > 0} A(\kappa) > A(0) \quad (24)$$

if and only if

$$\max\left(\frac{\sigma_1}{\sigma_2}, \frac{\sigma_2}{\sigma_1}\right) > 3 + 2\sqrt{2}. \quad (25)$$

If the inequality is satisfied, there exist exactly two positive crossings $0 < \kappa_1 < \kappa_2$ such that $A(\kappa) > A(0) \iff \kappa_1 < \kappa < \kappa_2$, and

$$\kappa_1 \kappa_2 = 1. \quad (26)$$

Proof. The homogeneous case $r = 1$ is handled separately and gives no enhancement. Assume therefore $r > 1$. Then $(r^2 - 1)^2 > 0$, so the sign of the discriminant in Eq. (21) is exactly the sign of $r^4 - 34r^2 + 1$.

Let $x = r^2$. The equation $x^2 - 34x + 1 = 0$ has roots $x_{\pm} = 17 \pm 12\sqrt{2}$. Because $r > 1$, the relevant branch is $r^2 > 17 + 12\sqrt{2}$. But $(3 + 2\sqrt{2})^2 = 17 + 12\sqrt{2}$, so $r > 3 + 2\sqrt{2}$.

For such r , the coefficient $r^4 - 18r^2 + 1$ is positive and $B_A(\kappa; r)$ is a downward-opening quadratic with two positive roots. Because the leading and constant coefficients of B_A are both $-8r^2$, Vieta's formula gives

$$\kappa_1 \kappa_2 = \frac{-8r^2}{-8r^2} = 1.$$

The quadratic is positive precisely between its two roots, completing the proof. $\square$

At the boundary $r = 3 + 2\sqrt{2}$ the two crossings merge into a tangency and $A(\kappa) \leq A(0)$ for all κ . By Eq. (26) the merged root is $\kappa = 1$; explicitly,

$$B_A(\kappa; r_A^*) = -8(r_A^*)^2(\kappa - 1)^2, \quad (27)$$

so the tangency occurs at $\kappa = 1$ and nowhere else. This coupling is used in Section 11.

Corollary 1 (Homogeneous and subthreshold systems). *If $1 \leq r \leq 3 + 2\sqrt{2}$, then $A(\kappa) \leq A(0)$ for every $\kappa \geq 0$.*

9 A Separate Threshold for Residual Distinction

The conditional residual variance has a distinct enhancement condition. For the same equal-relaxation normalization, $D(0) = \frac{1}{2}$, and direct simplification gives

$$D(\kappa) - D(0) = \frac{\kappa[-2(r^2 + 1)\kappa^2 + (r^4 - 8r^2 - 1)\kappa - 4r^2]}{4(\kappa + 1)(\kappa^2 r^2 + \kappa^2 + 4\kappa r^2 + 2r^2)}. \quad (28)$$

The sign-controlling quadratic is

$$B_D(\kappa; r) = -2(r^2 + 1)\kappa^2 + (r^4 - 8r^2 - 1)\kappa - 4r^2, \quad (29)$$

whose discriminant factors exactly as

$$\Delta_D = (r^2 - 1)^2(r^2 - 4r + 1)(r^2 + 4r + 1). \quad (30)$$

Proposition 5 (Residual-variance enhancement threshold). *For $r \geq 1$,*

$$\exists \kappa > 0 : D(\kappa) > D(0) \quad (31)$$

if and only if $r > 2 + \sqrt{3}$.

Proof. For $r > 1$ the factors $(r^2 - 1)^2$ and $r^2 + 4r + 1$ are strictly positive. Therefore real crossings require $r^2 - 4r + 1 > 0$. For $r \geq 1$ this holds precisely when $r > 2 + \sqrt{3}$. Above this threshold the linear coefficient $r^4 - 8r^2 - 1$ is positive and the downward-opening quadratic B_D becomes positive on a nonempty interval of positive κ . $\square$

Hence

$$2 + \sqrt{3} < 3 + 2\sqrt{2}, \quad (32)$$

numerically $2 + \sqrt{3} \approx 3.73205$ against $3 + 2\sqrt{2} \approx 5.82843$. Thus residual distinction can become enhanced at heterogeneity levels too small to enhance the total stationary covariance volume. This supplies an exact example of a central methodological point: different legitimate measures of relational structure need not identify the same threshold.

Remark 2. Unlike the volume crossings, the two residual crossings do *not* satisfy a reciprocal identity: the leading and constant coefficients of B_D are $-2(r^2 + 1)$ and $-4r^2$, so $\kappa_1^D \kappa_2^D = 2r^2/(r^2 + 1)$, which equals 1 only at $r = 1$. The reciprocal structure that the two thresholds *do* share is in the heterogeneity variable, not the coupling variable; that is the subject of the next section.

10 Exact Monotonicity of Coherence

We now prove, rather than numerically assume, the monotonicity of the Gaussian correlation. For equal relaxation rates,

$$\rho^2(\kappa) = \frac{\kappa^2(\kappa + 1)^2(r^2 + 1)^2}{(\kappa^2 r^2 + \kappa^2 + 4\kappa + 2)(\kappa^2 r^2 + \kappa^2 + 4\kappa r^2 + 2r^2)}. \quad (33)$$

Proposition 6 (Strict monotonicity of coherence). *For every $r > 0$ and $\kappa > 0$, $\frac{d}{d\kappa}\rho^2(\kappa) > 0$. Consequently $\rho(\kappa)$ and $C(\kappa) = -\frac{1}{2}\log(1 - \rho^2)$ are strictly increasing functions of κ .*

Proof. Differentiating Eq. (33) and factoring gives

$$\frac{d\rho^2}{d\kappa} = \frac{2\kappa(\kappa+1)(r^2+1)^2 P(\kappa, r)}{(\kappa^2 r^2 + \kappa^2 + 4\kappa + 2)^2 (\kappa^2 r^2 + \kappa^2 + 4\kappa r^2 + 2r^2)^2}, \quad (34)$$

where

$$P(\kappa, r) = \kappa^4(r^2+1)^2 + 16\kappa^3 r^2 + 24\kappa^2 r^2 + 16\kappa r^2 + 4r^2. \quad (35)$$

Every factor in the numerator and denominator is strictly positive for $\kappa > 0$, $r > 0$. Therefore $d\rho^2/d\kappa > 0$. Because $\rho \geq 0$ and $x \mapsto -\frac{1}{2} \log(1-x)$ is strictly increasing for $0 \leq x < 1$, both ρ and C are strictly increasing. $\square$

11 Reciprocal Structure of the Two Thresholds

This section is new in version 2. It records that the two thresholds of Sections 8 and 9, which version 1 presented only as two unrelated algebraic numbers, are two instances of one structure, and that they meet at a single identifiable coupling.

11.1 Both thresholds are reciprocal units

Write $s(x) = x + 1/x$ as in Eq. (16). If $x > 0$ satisfies $x^2 - cx + 1 = 0$ then $s(x) = c$, and the two roots of that quadratic are x and $1/x$. Both thresholds are of this form:

$$(r_A^*)^2 - 6r_A^* + 1 = 0, \quad (r_D^*)^2 - 4r_D^* + 1 = 0, \quad (36)$$

so that

$$s(r_A^*) = r_A^* + \frac{1}{r_A^*} = 6, \quad s(r_D^*) = r_D^* + \frac{1}{r_D^*} = 4. \quad (37)$$

The invariance of s under $x \mapsto 1/x$ is not decorative: it is exactly the relabelling symmetry $X \leftrightarrow Y$, under which $r \mapsto 1/r$ and every quantity in this paper is unchanged. The threshold conditions of Theorem 1 and Proposition 5 can therefore be written in a manifestly symmetric form that needs no $\max(\cdot, \cdot)$:

$$\sup_{\kappa > 0} A(\kappa) > A(0) \iff s(r) > 6, \quad \exists \kappa > 0 : D(\kappa) > D(0) \iff s(r) > 4. \quad (38)$$

In the variance variable $t = r^2$, using $w(t) = s(r)^2 - 2$, the same two conditions read $w(t) > 34$ and $w(t) > 14$. The number 34 is the coefficient already visible in the discriminant factor $r^4 - 34r^2 + 1$ of Eq. (21); that factor is nothing but $t^2 - 34t + 1$, the quadratic whose reciprocal invariant is 34.

11.2 A cross-threshold identity at the merger

At $r = r_A^*$ the enhancement window has just closed, and by Eq. (27) it closes at $\kappa = 1$. We evaluate the shape of Σ there.

For $a_1 = a_2 = 1$, $\sigma_1 = 1$, $\sigma_2 = r$ and $\kappa = 1$, Eqs. (10)–(12) give

$$\text{tr } \Sigma(1) = \frac{1+r^2}{3}, \quad \det \Sigma(1) = \frac{r^4 + 14r^2 + 1}{192}. \quad (39)$$

Consider the scale-free shape statistic

$$J_2(\Sigma) = \frac{(\text{tr } \Sigma)^2}{\det \Sigma}, \quad (40)$$

which for a positive-definite 2×2 matrix with eigenvalues $\mu_1 \geq \mu_2 > 0$ satisfies

$$J_2 = \frac{(\mu_1 + \mu_2)^2}{\mu_1 \mu_2} = q + \frac{1}{q} + 2 = s(q) + 2, \quad q = \frac{\mu_1}{\mu_2}. \quad (41)$$

J_2 is a monotone function of the eigenvalue ratio, so it determines the ratio uniquely; it is the trace/determinant sphericity statistic in the sense of Mauchly [16] (see also [17, 24]). Substituting Eq. (39) into Eq. (40) and writing the result in terms of $s = s(r)$ gives, for every $r > 0$, the closed form

$$J_2(\Sigma(1)) = \frac{64}{3} \cdot \frac{s^2}{s^2 + 12}. \quad (42)$$

Theorem 2 (Cross-threshold identity). *Let $r = r_A^* = 3 + 2\sqrt{2}$ and $\kappa = 1$, the unique coupling at which the volume enhancement region degenerates. Then the stationary covariance matrix satisfies*

$$J_2(\Sigma(1)) = 16, \quad \text{equivalently} \quad q + \frac{1}{q} = 14, \quad (43)$$

and therefore its eigenvalue ratio is exactly

$$q = \frac{\mu_1}{\mu_2} = 7 + 4\sqrt{3} = (2 + \sqrt{3})^2 = (r_D^*)^2. \quad (44)$$

Proof. Put $s = s(r_A^*) = 6$ in Eq. (42): $\frac{64}{3} \cdot 36/48 = 16$. By Eq. (41), $s(q) = J_2 - 2 = 14$, so q satisfies $q^2 - 14q + 1 = 0$; the root with $q > 1$ is $7 + 4\sqrt{3}$. Finally $(2 + \sqrt{3})^2 = 7 + 4\sqrt{3}$, and $\Sigma(1)$ has two distinct positive eigenvalues because $(\text{tr } \Sigma)^2 - 4 \det \Sigma = 12 \det \Sigma > 0$. $\square$

So the two thresholds are linked by an exact identity: *the covariance eigenvalue ratio at the critical coupling of the volume threshold is the square of the residual threshold.* In the invariant variables of Eq. (16) the chain is

$$s(r_A^*) = 6 \longrightarrow J_2 = \frac{64}{3} \cdot \frac{36}{48} = 16 \longrightarrow s(q) = 14 = w((r_D^*)^2) \longrightarrow q = (r_D^*)^2.$$

An equivalent form is additive: writing $\eta = \frac{1}{2} \log q$ for the log-anisotropy of the covariance ellipse,

$$\eta = \log r_D^* = \text{arcosh } 2. \quad (45)$$

Remark 3 (What this identity is and is not). Theorem 2 is an exact statement about this model, and it is proved by substitution; we make no claim that it reflects a general principle. Two honest caveats. First, the numerical coincidence is assisted by the normalization $a_1 = a_2 = 1$: J_2 is scale-free in Σ , but Eq. (42) is specific to the equal-relaxation family. Second, the appearance of r_D^* in Eq. (44) is a statement about the *number* $(2 + \sqrt{3})^2$; we have not exhibited a mechanism that derives the residual threshold from the volume-critical covariance geometry, and we do not assert that one exists. What the identity does establish is that the two thresholds are not independent algebraic accidents: both are reciprocal units, and the volume-critical covariance realizes the residual unit as its anisotropy. Both statements, and Eq. (45), are machine-checked (Section 15).

Remark 4 (Failure beyond two nodes). The shape statistic J_2 determines the eigenvalue ratio only for 2×2 matrices. For $n \geq 3$ the pair $(\text{tr } \Sigma, \det \Sigma)$ no longer fixes the spectrum's shape: the diagonal matrices with spectra $(1, 8, 12)$ and $(2, 3, 16)$ have the same trace 21 and the same determinant 96 but different extreme-eigenvalue ratios, 12 and 8. Any extension of Theorem 2 to larger networks must therefore use more than these two invariants. This counterexample is also machine-checked.

Table 2: Representative values for $a_1 = a_2 = 1$, $\sigma_1 = 1$, $\sigma_2 = 10$.

κ	ρ	$C(\kappa)$	$D(\kappa)$	$A(\kappa)/A(0)$
0	0.0000	0.0000	0.5000	1.0000
0.5	0.6792	0.3093	1.3131	1.8613
1.0	0.7376	0.3926	2.0330	2.3752
1.41819	0.7658	0.4415	2.3493	2.4574
2.0	0.7949	0.4996	2.5406	2.3780
5.0	0.8734	0.7196	2.2877	1.6378
10.0	0.9225	0.9522	1.6346	1.0119
100	0.9903	1.9728	0.2406	0.1245

12 Representative Numerical Example

Take $a_1 = a_2 = 1$, $\sigma_1 = 1$, $\sigma_2 = 10$. Then $r = 10 > 3 + 2\sqrt{2}$ and the uncoupled volume is $A(0) = 25$. The first crossing occurs at approximately $\kappa_1 = 0.0984954$, the second at $\kappa_2 = 10.1527546$, with $\kappa_1\kappa_2 = 1$ to numerical precision. The global maximum occurs at approximately $\kappa^* = 1.41819$, where $A(\kappa^*)/A(0) \approx 2.4574$. Representative values are shown in Table 2.

The sequence is therefore

uncoupled $\rightarrow$ initial compression $\rightarrow$ intermediate expansion
 $\rightarrow$ increasing redundancy $\rightarrow$ strong-coupling collapse.

13 Figures

The accompanying reproducibility script generates four figures.

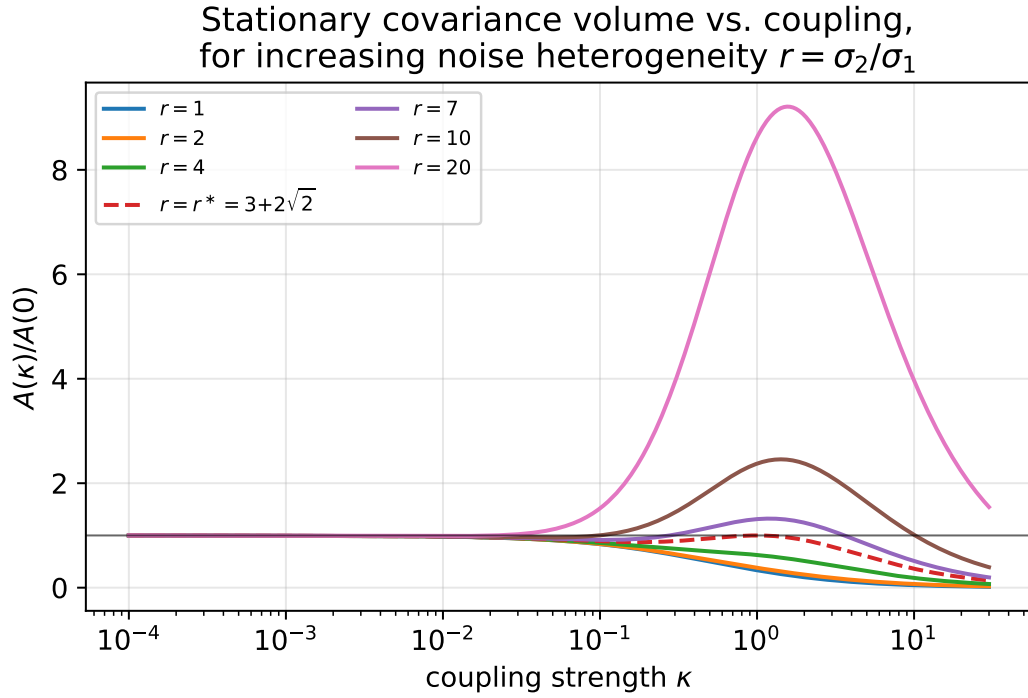


Figure 1: Stationary covariance-volume ratio $A(\kappa)/A(0)$ for several values of the noise-amplitude ratio r . The enhancement regime appears only above $r = 3 + 2\sqrt{2}$.

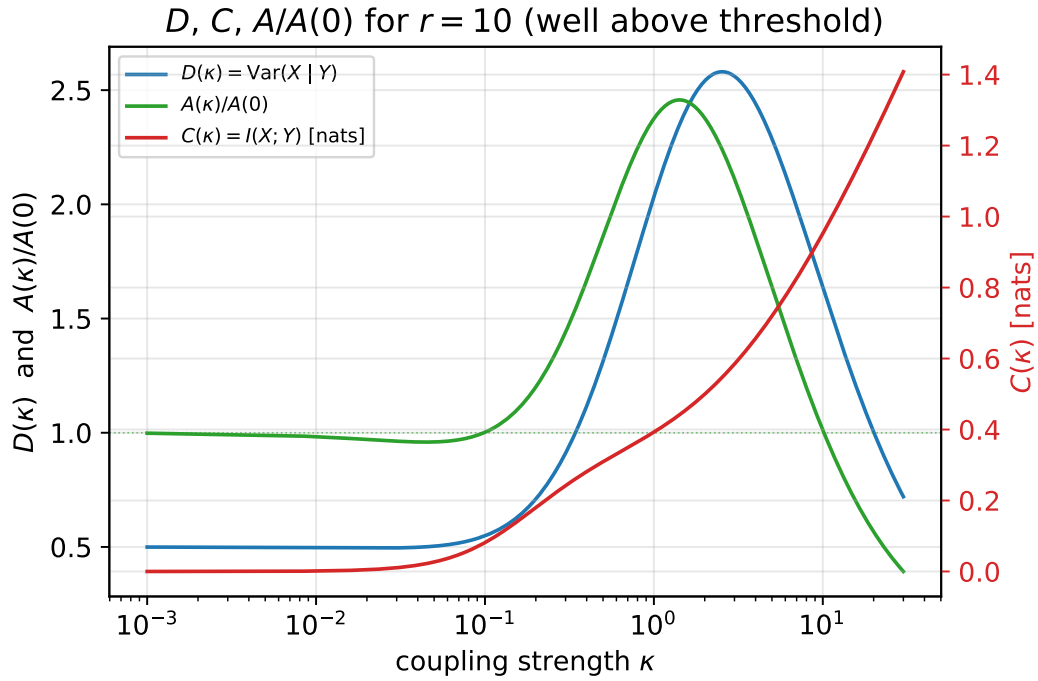


Figure 2: $D(\kappa)$, $C(\kappa)$, and $A(\kappa)/A(0)$ for $r = 10$. Coherence increases monotonically while the other two measures are non-monotonic.

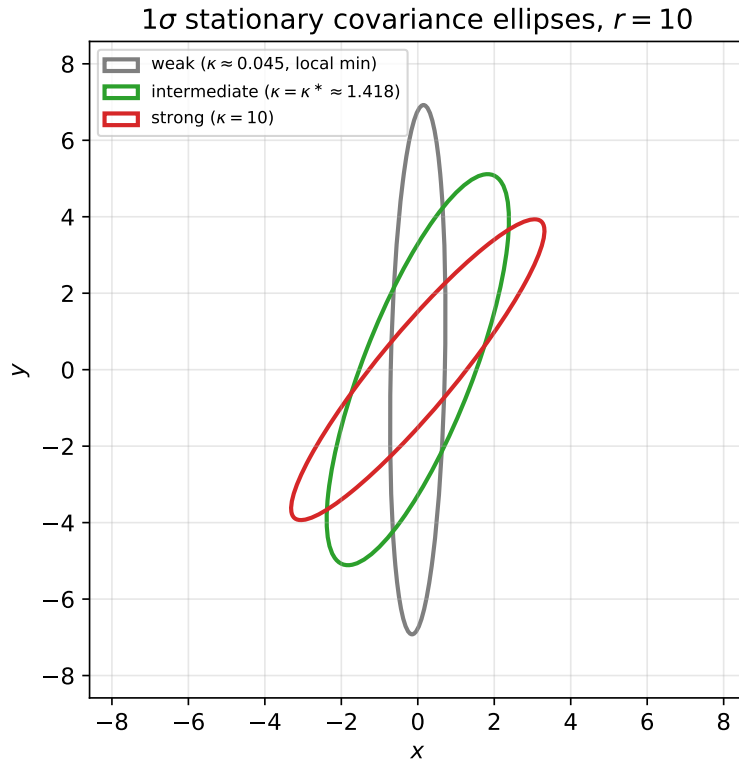


Figure 3: Stationary covariance ellipses for representative weak, intermediate, and strong coupling.

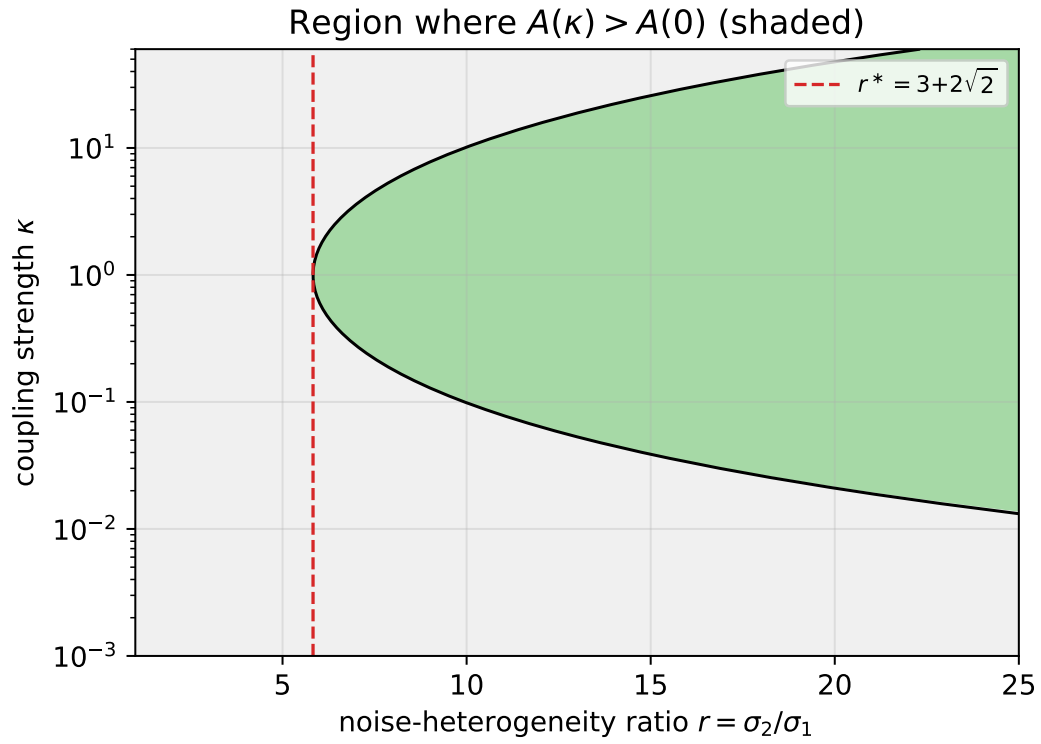


Figure 4: Region in the (r, κ) plane for which $A(\kappa) > A(0)$. The enhancement region opens at the exact threshold $r = 3 + 2\sqrt{2}$.

14 Strong-Coupling Limit of the Covariance Geometry

For $a_1 = a_2 = 1$, $\sigma_1 = 1$, $\sigma_2 = r$, the covariance components reduce to

$$\Sigma_{xx} = \frac{\kappa^2 r^2 + \kappa^2 + 4\kappa + 2}{4(2\kappa^2 + 3\kappa + 1)}, \quad (46)$$

$$\Sigma_{yy} = \frac{\kappa^2 r^2 + \kappa^2 + 4\kappa r^2 + 2r^2}{4(2\kappa^2 + 3\kappa + 1)}, \quad (47)$$

$$\Sigma_{xy} = \frac{\kappa(r^2 + 1)}{4(2\kappa + 1)}. \quad (48)$$

Proposition 7 (Strong-coupling asymptotics). *As $\kappa \rightarrow \infty$,*

$$1 - \rho^2(\kappa) \sim \frac{2}{\kappa}, \quad D(\kappa) \sim \frac{r^2 + 1}{4\kappa}, \quad A(\kappa) \sim \frac{(r^2 + 1)^2}{32\kappa}, \quad (49)$$

while $\Sigma_{xx}, \Sigma_{yy}, \Sigma_{xy} \rightarrow \frac{r^2 + 1}{8}$.

Thus $x - y$ becomes increasingly suppressed and the covariance ellipse collapses toward the line $x = y$. In the modal form of Eq. (23) this is transparent: the longitudinal variance $\tilde{\Sigma}_{11} = (1 + r^2)/4$ is independent of κ , while the transverse variance $\tilde{\Sigma}_{22}$ decays like $1/(2\kappa)$ and the modal correlation like $1/\kappa$.

Remark 5 (On the word “synchronization”). The title and the phrase “synchronization collapse” need to be read in a restricted sense, and version 1 was not explicit enough about this. Three points.

First, the collapse described here is a property of the *stationary covariance ellipse*, not of trajectories. The full mathematical state space remains two-dimensional for every finite κ , the stationary Gaussian retains full support on $\mathbb{R}^2$ for every finite κ , and $x(t) \neq y(t)$ almost surely at every time. Nothing here is synchronization in the sense of trajectories converging.

Second, the limit is singular: $\Sigma(\kappa)$ becomes rank-deficient only as $\kappa \rightarrow \infty$. “Dimensional collapse” refers to that limit, and to the fact that the variance transverse to $x = y$ vanishes at rate $1/\kappa$, not to a literal reduction of state-space dimension at any finite coupling.

Third, the relation to the master-stability literature [18] is geometric only. That framework asks whether a synchronization manifold of a nonlinear system is transversally stable; here the linear system has a globally attracting stationary distribution for every $\kappa \geq 0$, there is no stability transition, and the transverse direction is always stable. What is shared is the decomposition into a longitudinal and a transverse mode relative to $x = y$ — and, as Section 7.1 shows, even that decomposition behaves differently here, because heterogeneous noise correlates the two modes.

A more literal description of the phenomenon would be *strong-coupling degeneration of the stationary covariance toward the diagonal*. We retain the shorter phrase in the title, for continuity with version 1, with this remark as its definition.

15 Machine-Checked Verification

The central algebraic content of this paper has been formalized and machine-checked in Lean 4 with mathlib. The development is public at <https://github.com/innerlightr-wq/ou-threshold-lean>; the state corresponding to this version of the paper is recorded there as frozen snapshots under `verified/`, each with its own toolchain file, dependency manifest, SHA-256 manifest, and build log.

What is formalized. The formalization works at the level of *exact real algebra*: the stationary covariance is *defined* by the closed forms (10)–(12) in the equal-relaxation normalization, and the Lyapunov equation is then *proved* to hold for it. Specifically, the following are machine-checked:

1. $M\Sigma + \Sigma M^\top = -Q$ for the stated Σ , for all $\kappa \geq 0$ and all r (`cov_lyapunov`), together with uniqueness of the solution (`lyapunov_unique`) and $\det \Sigma = A(\kappa)$ (`det_cov`).
2. Theorem 1, in the form $(\exists \kappa > 0, A(\kappa, r) > A(0, r)) \iff r > 3 + 2\sqrt{2}$ for $r \geq 1$ (`exists_volume_improvement_iff_threshold`), including the discriminant factorization (21).
3. Proposition 5, in the form $(\exists \kappa > 0, D(\kappa, r) > D(0, r)) \iff r > 2 + \sqrt{3}$ for $r \geq 1$ (`exists_residual_improvement_iff_threshold`), including the factorization (30).
4. The tangency (27) at $r = r_A^*$ with double root $\kappa = 1$, and $A(\kappa, r_A^*) \leq A(0, r_A^*)$ for all $\kappa \geq 0$ (Corollary 1 at the boundary).
5. The reciprocal relations (37) and $w(r^2) = s(r)^2 - 2$, and Theorem 2: the closed form (42), the value $J_2 = 16$, the eigenvalue ratio $q = 7 + 4\sqrt{3} = (r_D^*)^2$, the non-vacuity of that statement (the matrix really has two distinct real eigenvalues), and the rapidity form (45).
6. The $n \geq 3$ counterexample (1, 8, 12) versus (2, 3, 16).

Each statement is checked by Lean’s kernel and depends only on the three standard axioms of `mathlib`’s classical foundation (`propext`, `Classical.choice`, `Quot.sound`); `#print axioms` output is included in the snapshots. There are no `sorry`s and no appeals to `native_decide`.

What is not formalized. This is a companion to the paper, not a formalization of it, and the gap should be stated plainly. Not formalized are: the stochastic-process content of Proposition 1 (existence and uniqueness of the stationary Gaussian measure for a Hurwitz linear SDE, which is quoted from the literature and used to justify the Lyapunov equation, not derived in Lean); the general-rate covariance (13) and Proposition 3, which are proved here by hand and checked with computer algebra only; the monotonicity of coherence (Proposition 6); the strong-coupling asymptotics (Proposition 7); the numerical table; and the figures. Formalizing Proposition 1 would require a treatment of linear SDEs that, to our knowledge, `mathlib` does not currently have.

In short: the machine-checked part is exactly the algebra that carries the two thresholds and the identity between them. The modelling step — that this Σ is the stationary covariance of that stochastic process — is classical, and is assumed rather than verified.

16 Relation to the Dissipative Bosonic Josephson Junction

An earlier study of a phenomenologically damped bosonic Josephson junction found an exact separation between local contraction information and global basin accessibility [6]. In that model $\text{tr } J = -\gamma$ throughout the interior phase space, so the local phase-space contraction rate is independent of the interaction parameter Λ , while the measured global trapped-basin fraction varies substantially with Λ . The conservative system also exhibits two different robustness measures with different qualitative behavior: the energy separation is monotonic while the directional separatrix width possesses an interior maximum.

The present OU calculation is not dynamically equivalent to that system. The common lesson is methodological:

Exact local information can be insufficient to determine global parameter-dependent geometry, and distinct scalar measures of the same system need not identify the same structural threshold.

The present paper strengthens the second part of that observation by producing two exact heterogeneity thresholds, $r_D^* = 2 + \sqrt{3}$ for conditional residual variance and $r_A^* = 3 + 2\sqrt{2}$ for stationary covariance volume — and, in Theorem 2, by showing that in this model the two are nonetheless algebraically related.

17 A Provisional Intermediate Interaction Regime

For chosen thresholds $d_0, c_0 > 0$, one could define

$$\mathcal{C} = \{\kappa > 0 : A(\kappa) > A(0), D(\kappa) \geq d_0, C(\kappa) \geq c_0\}. \quad (50)$$

Such a set can be interpreted as an intermediate interaction regime satisfying simultaneous requirements of stationary volume enhancement, retained residual structure, and nonzero coherence. We do not elevate this definition to a theorem because d_0 and c_0 are externally chosen.

The exact results requiring no arbitrary thresholds are instead $r_D^* = 2 + \sqrt{3}$, $r_A^* = 3 + 2\sqrt{2}$, the identity $\kappa_1 \kappa_2 = 1$, Theorem 2, and the strict monotonicity of $C(\kappa)$.

18 Limitations

Several limitations are essential. Items 1–7 restate version 1; items 8–11 are new.

1. The model is linear, Gaussian, two-dimensional, and driven by independent additive white noise.
2. The principal volume threshold $3 + 2\sqrt{2}$ is proved only for equal intrinsic relaxation rates. Proposition 3 is the only general-rate result in the paper.
3. The residual threshold $2 + \sqrt{3}$ is likewise derived in the equal-relaxation normalization.
4. The stationary distribution has full support in $\mathbb{R}^2$ for every finite κ , so $A(\kappa)$ measures covariance volume rather than mathematical reachability or support.
5. The model possesses a unique stationary state and therefore does not address multistable basin geometry.
6. The threshold values are properties of this model. No universal physical significance is attributed to either algebraic constant.
7. No inference is made from these calculations to consciousness, artificial intelligence, biological evolution, social systems, or cosmology.
8. The word “synchronization” is used in the restricted sense of Remark 5: degeneration of the stationary covariance toward the diagonal in a singular limit. No trajectory-level or stability-theoretic synchronization statement is made or implied, and the master-stability framework is not applicable to this system.
9. “Expansion” means an increase of $\det \Sigma$ relative to its uncoupled value at the same parameters. It is a statement about second moments of a Gaussian measure with fixed full support, and it is bounded: by Proposition 7, $A(\kappa) \rightarrow 0$, so any enhancement is a transient of the coupling parameter, not a persistent gain.

10. Theorem 2 is an exact identity, verified symbolically and mechanically, but it is an identity within one two-parameter family. We have not derived it from a structural principle, we do not claim it generalizes, and the $n \geq 3$ counterexample above shows that the invariant it is built from does not by itself survive to larger networks.
11. The formalization of Section 15 verifies algebra, not modelling. In particular, no part of the stochastic-process argument is machine-checked, and a reader who doubts Proposition 1 gains nothing from the Lean development.

19 Nonlinear Bistable Extension

A natural falsification test is a pair of coupled stochastic bistable systems:

$$dx = -V_1'(x) dt + \kappa(y - x) dt + \sigma_1 dW_1, \quad (51)$$

$$dy = -V_2'(y) dt + \kappa(x - y) dt + \sigma_2 dW_2, \quad (52)$$

with $V_i(x) = \frac{x^4}{4} - \frac{a_i x^2}{2}$.

Each uncoupled subsystem possesses two metastable wells, giving four coarse joint configurations: $(--)$, $(-+)$, $(+-)$, $(++)$. This system can test whether an intermediate coupling regime preserves mixed configurations while modifying transition structure, and whether strong coupling suppresses the mixed states in favor of synchronized configurations. No results for this nonlinear extension are reported here.

20 Companion Note: A General Two-Mode Reduction

The volume threshold of Section 8 turns out not to depend on the graph at all, but only on a two-dimensional reduction, and a machine-checked account of this is included in the companion repository (module `ActivePlane`). Briefly: let a Hurwitz drift have a two-dimensional invariant plane with decay rates $D_1, D_2 > 0$, and let the forcing be a rank-one deformation of the isotropic one, $Q = q_b(I + (\tau - 1)ee^\top)$, where e is a unit *excitation direction* (the direction along which the forcing is anisotropic; e here is never Euler's number) making an angle with the drift eigenbasis given by $e = cu + sv$, $c^2 + s^2 = 1$. Then the stationary covariance in that plane satisfies the palindromic determinant identity $4D_1D_2 \det \Sigma = q_b^2(\mathcal{A}\tau^2 + (1 - 2\mathcal{A})\tau + \mathcal{A})$ with $\mathcal{A} = c^2s^2\nu^2$ and $\nu = (D_1 - D_2)/(D_1 + D_2)$; dividing by τ gives $1 + \mathcal{A}(w(\tau) - 2)$ with w as in Eq. (16). The palindromic structure is the self-duality $Q_e(\tau) = \tau Q_{e^\perp}(1/\tau)$, which holds because $ee^\top + e^\perp e^{\perp\top} = I$; this is why a reciprocal invariant appears at all, and it explains why the relevant reciprocal variable is the *variance* ratio $\tau = r^2$ rather than the amplitude ratio. The present paper's threshold is the instance $\tau = r^2$, $w = 34$, of that identity. A separate write-up of the general statement is in preparation and is not part of this version.

21 Strongest Defensible Contribution

The principal exact contribution can be summarized as follows.

For two diffusively coupled Ornstein–Uhlenbeck processes with equal relaxation rate, the stationary covariance volume $A(\kappa) = \det \Sigma(\kappa)$ always satisfies $A'(0) < 0$. Nevertheless, $\sup_{\kappa > 0} A(\kappa) > A(0)$ if and only if $\max(\sigma_1/\sigma_2, \sigma_2/\sigma_1) > 3 + 2\sqrt{2}$. When the enhancement

regime exists, its two boundary couplings satisfy $\kappa_1\kappa_2 = 1$. Separately, the conditional residual variance can exceed its uncoupled value if and only if the noise-amplitude ratio exceeds $2 + \sqrt{3}$. Gaussian coherence increases strictly with coupling for every positive noise ratio. Both thresholds are reciprocal units, $s(r_A^*) = 6$ and $s(r_D^*) = 4$; and at the coupling $\kappa = 1$ where the volume enhancement region degenerates, the covariance eigenvalue ratio is exactly $(r_D^*)^2 = 7 + 4\sqrt{3}$.

The two distinct thresholds demonstrate exactly that covariance-volume expansion, retained residual distinction, and coherence are not interchangeable measures, even though all three arise from the same covariance matrix. That they are nonetheless algebraically related, in the precise sense of Theorem 2, is the main new result of this version.

22 Conclusion

Two diffusively coupled OU processes provide a minimal analytically solvable setting in which heterogeneity, coupling, coherence, and covariance geometry can be separated exactly.

The first result is universal: $A'(0) < 0$. Infinitesimal coupling always contracts stationary covariance volume.

The global response, however, depends on heterogeneity. In the equal-relaxation model, $\exists \kappa > 0 : A(\kappa) > A(0)$ if and only if $r > 3 + 2\sqrt{2}$. A distinct and lower threshold, $r > 2 + \sqrt{3}$, governs enhancement of conditional residual variance. Both are reciprocal units, and they meet at the critical coupling in the sense of Theorem 2. Meanwhile $C(\kappa)$ increases monotonically for all positive r , while sufficiently strong coupling drives $\rho \rightarrow 1$, $D \rightarrow 0$, $A \rightarrow 0$.

The resulting picture is therefore not that coupling is generically enriching. Rather, sufficiently heterogeneous systems in this particular model can exhibit a bounded intermediate range in which stationary covariance volume exceeds its uncoupled value before the stationary covariance ultimately degenerates toward the diagonal. Whether an analogous architecture survives in nonlinear and multistable systems remains an open question.

Acknowledgments

AI assistance (Anthropic’s Claude, used through Claude Code) was used for symbolic algebra verification, numerical computation, literature search and citation verification, construction and checking of the Lean 4 formalization reported in Section 15, and manuscript preparation, including the reconstruction of this document from the version 1 PDF described in Section 0. Every mathematical statement retained from version 1 was re-derived symbolically before being carried over, and the statements listed in Section 15 are in addition checked by Lean’s kernel, which does not depend on the assistant’s correctness. All conceptual development, research direction, interpretation, and final responsibility for the work remain with the author.

A Reproducibility

All symbolic derivations and numerical results were independently checked using Python with SymPy, NumPy, and SciPy. The explicit stationary covariance solution was compared with the independent numerical solver `scipy.linalg.solve_continuous_lyapunov` at multiple parameter points. The maximum observed elementwise discrepancy was approximately 2.8×10^{-14} .

The reproducibility script accompanying this manuscript verifies:

1. the exact Lyapunov solution;
2. Proposition 3;
3. the difference formula, Eq. (19);
4. the discriminant factorization, Eq. (21);
5. the threshold $3 + 2\sqrt{2}$;
6. the reciprocal identity $\kappa_1\kappa_2 = 1$;
7. the residual-variance threshold $2 + \sqrt{3}$;
8. the exact positivity factorization for $d\rho^2/d\kappa$;
9. the strong-coupling asymptotics;
10. the numerical table and figures.

New in this version, and checked the same way before being stated: the modal form (23) and its identity with Eq. (18); the merger invariant (42); and the values in Table 1. These, together with the statements listed in Section 15, are in addition machine-checked in Lean 4.

B Note on the figures in this version

The figures of version 1 were reproduced in this version by extracting the four vector graphics from the deposited version 1 PDF, without modification of their content. They are therefore identical to the version 1 figures and were not regenerated. The script that originally produced them was not preserved with the manuscript source; see Section 0.

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